Proof for Theorem 1 (Main Theorem; Thm:generalized-Thinning)

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Key Lemmas

Lemma 1 (Pushforward of a two-stage sample under reconstruction)**.** *Let X and Y be measurable spaces. Let μ be a measure on X, let K assign to each $x \in X$ a measure K_x on Y, and let $T : Y \to X$. Assume:*

(i) *μ is a probability measure;*

(ii) *K is a Markov kernel, meaning that every K_x is a probability measure and $x \mapsto K_x(A)$ is measurable for every measurable $A \subseteq Y$;*

(iii) *T is measurable;*

(iv) *for μ-almost every $x \in X$, one has $T(y) = x$ for K_x-almost every $y \in Y$.*

For a measure ν on X, define

$$(\nu K)(A) = \int_X K_x(A)\, \nu(dx) \qquad (A \subseteq Y \text{ measurable}),$$

and

$$(\nu \boxtimes K)(S) = \int_X K_x\big(\{y \in Y : (x, y) \in S\}\big)\, \nu(dx) \qquad (S \subseteq X \times Y \text{ measurable}).$$

Also write $f_{\#}\nu$ for the pushforward of ν by a measurable map f. Then, with $F(y) = (T(y), y)$,

$$F_{\#}(\mu K) = \big(T_{\#}(\mu K)\big) \boxtimes K.$$

Proof. Set $F(y) = (T(y), y)$. By Condition (iii), F is measurable. Condition (iv) says that for μ-almost every $x \in X$, the equality $T(y) = x$ holds for K_x-almost every $y \in Y$. Hence, for μ-almost every x, the maps

$$F(y) = (T(y), y) \qquad \text{and} \qquad F_x(y) = (x, y)$$

agree K_x-almost everywhere. Consequently,

$$(K_x)_{\#}F = (K_x)_{\#}F_x \qquad \text{for } \mu\text{-almost every } x. \tag{1}$$

Indeed, for any measurable set $U \subseteq X \times Y$, the two preimages of U under F and F_x differ only by a K_x-null set.

Since K is a Markov kernel, the map $x \mapsto K_x$ is μ-almost everywhere measurable, so the standard formulas for the law of a two-stage sample apply.

We first prove

$$F_{\#}(\mu K) = \mu \boxtimes K. \tag{2}$$

By extensionality of measures, it suffices to evaluate both sides on an arbitrary measurable set $S \subseteq X \times Y$. Using the definition of pushforward and of μK,

$$F_{\#}(\mu K)(S) = (\mu K)(F^{-1}S) = \int_X K_x(F^{-1}S)\, \mu(dx).$$

On the other hand, by the definition of $\mu \boxtimes K$,

$$(\mu \boxtimes K)(S) = \int_X K_x\big(\{y \in Y : (x, y) \in S\}\big)\, \mu(dx).$$

For μ-almost every x, equality (1) gives

$$K_x(F^{-1}S) = ((K_x)_{\#}F)(S) = ((K_x)_{\#}F_x)(S) = K_x(F_x^{-1}S) = K_x\big(\{y \in Y : (x, y) \in S\}\big).$$

Thus the two integrands agree for μ-almost every x, so the integrals are equal. Since this holds for every measurable S, equality (2) follows.

Next we prove

$$T_{\#}(\mu K) = \mu. \tag{3}$$

The field $y \mapsto \delta_{T(y)}$ is measurable with respect to μK, because T is measurable. Therefore

the pushforward by T can be written as the two-stage measure

$$T_{\#}(\mu K) = \int_Y \delta_{T(y)}\,(\mu K)(dy).$$

Using associativity of two-stage sampling,

$$T_{\#}(\mu K) = \int_X \left(\int_Y \delta_{T(y)}\, K_x(dy) \right) \mu(dx) = \int_X (K_x)_{\#} T\, \mu(dx). \tag{4}$$

Now Condition (iv) says that for μ-almost every x, the function T agrees with the constant map $y \mapsto x$ on a K_x-full set. Hence

$$(K_x)_{\#} T = (K_x)_{\#}(y \mapsto x) = \delta_x \qquad \text{for } \mu\text{-almost every } x. \tag{5}$$

The last equality uses that K_x is a probability measure, which is Condition (ii). Therefore the kernels $x \mapsto (K_x)_{\#} T$ and $x \mapsto \delta_x$ agree μ-almost everywhere, so replacing the first by the second in (4) is valid. Consequently,

$$T_{\#}(\mu K) = \int_X \delta_x\, \mu(dx).$$

For every measurable $A \subseteq X$, the right-hand side equals

$$\int_X 1_A(x)\, \mu(dx) = \mu(A),$$

so it is the pushforward of μ by the identity map. Thus $T_{\#}(\mu K) = \mu$, proving (3).

Combining (2) and (3), we obtain

$$F_{\#}(\mu K) = \mu \boxtimes K = \big(T_{\#}(\mu K)\big) \boxtimes K.$$

This is exactly the required identity.

□

Main Theorem

Theorem 2 (Generalized thinning representation)**.** *Let $K \geq 1$ be an integer, let X be a measurable space, let Ω be a nonempty parameter set, and let $T : X^K \to X$ be measurable. For each $\theta \in \Omega$, let P_θ be a probability measure on X. Let G be a Markov kernel from X to X^K, and for $x \in X$ write G_x for the corresponding probability measure on X^K. For*

$\theta \in \Omega$, let Q_θ be the probability measure on X^K obtained by sampling $X \sim P_\theta$ and then $Y \mid X = x \sim G_x$. For a probability measure η on X and a Markov kernel κ from X to X^K, write $\eta \otimes \kappa$ for the joint law of (Z, Y) when $Z \sim \eta$ and $Y \mid Z = z \sim \kappa_z$. Assume:

(i) for every θ, the coordinate maps $y \mapsto y_i$, $i = 1, \dots, K$, are mutually independent under Q_θ;

(ii) for every θ, for P_θ-almost every $x \in X$, for G_x-almost every $y \in X^K$, one has $T(y) = x$.

Then there exists a Markov kernel κ from X to X^K such that for every θ,

$$(T, \mathrm{id})_* Q_\theta = (T_* Q_\theta) \otimes \kappa.$$

Moreover, the same identity holds with $\kappa = G$:

$$(T, \mathrm{id})_* Q_\theta = (T_* Q_\theta) \otimes G$$

for every θ.

Proof. Fix $\theta \in \Omega$, and put $\mu = P_\theta$, $Q = Q_\theta$. Define

$$F : X^K \to X \times X^K, \qquad F(y) = (T(y), y).$$

Since T is measurable, F is measurable. By Condition (ii), for μ-almost every $x \in X$ we have $T(y) = x$ for G_x-almost every y, hence

$$F(y) = (T(y), y) = (x, y)$$

for G_x-almost every y. Since P_θ is a probability measure and G is a Markov kernel, Lemma 1 applies with $\mu = P_\theta$ and gives

$$F_* Q_\theta = (T_* Q_\theta) \otimes G.$$

Equivalently,

$$\mathrm{map}\big(y \mapsto (T(y), y)\big)\big((P_\theta) \,\mathrm{bind}\, G\big) = \mathrm{compProd}\big(\mathrm{map}\, T\big((P_\theta) \,\mathrm{bind}\, G\big)\big) G.$$

Since θ was arbitrary, this equality holds simultaneously for all $\theta \in \Omega$, with the same kernel G.

Now take $\kappa = G$. The kernel G is Markov by the initial hypotheses, so κ is a Markov

kernel. For every θ, the equality just established is precisely

$$(T, \mathrm{id})_* Q_\theta = (T_* Q_\theta) \otimes \kappa.$$

Thus the existential assertion holds. Reading the same equality with $\kappa = G$ gives the second required assertion. Hence both parts are proved.

□

Task outcomes and dependencies

Task registry snapshot: paper_fresh_20261028t224543381z_data_thinning, revision 6, captured 2026-10-28T23:30:04.121Z.

This report belongs to paper_fresh_20261028t224543381z_data_thinning (job paper_fresh_20261028t its own terminal report determines its outcome, since registry settlement can lag reporting.

Each task has its own outcome. A proved child does not certify the main task or discharge other external assumptions.